Universal Tangent-Kernel Closure in Two-Factor Linear Networks, and Its Failure Beyond

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Abstract

Finite-width gradient flow in a two-factor linear network admits an exact projection from factor space to the joint prediction–kernel state (f, K). For every scalar-output bias-free model

$$f = XW^\top a, \qquad G = XX^\top,$$

with arbitrary hidden width, input dimension, finite dataset, and differentiable loss, the answer is yes:

$$K = XW^\top WX^\top + \|a\|^2 G, \qquad \dot{K} = -(Gr)f^\top - f(Gr)^\top - 2(f^\top r)G.$$

The contribution is the closure statement and its consequences: the parameter representation disappears globally, the law is independent of the nontrivial balancing leaf $WW^\top - aa^\top$, and the kernel equation is driven by (f, r) rather than by K itself. Eliminating K gives an exact prediction-only Volterra equation with full, non-fading history and no remainder once K_0 is fixed. Thus the same dynamics are Markovian in (f, K) and memory-bearing in f alone.

For scalar deep-linear networks, we derive the finite hierarchy

$$\dot{e}_k = -2fr(d - k + 1)e_{k-1}, \qquad K = e_{d-1}, \qquad \dot{K} = -4fre_{d-2}.$$

At two factors, $e_{d-2} = 1$ and the universal closure holds. From three factors onward, regular fibers of (e_d, e_{d-1}) contain curves along which e_{d-2} varies, so no leaf-independent law $\dot{K} = \mathcal{F}(f, K, r)$ exists on the full positive state space. Conditioning on the conserved balancing leaf restores exact closure; at depth three, (f, K, I) is simply the complete symmetric state in invariant coordinates. The result is therefore a precise transition from universal driven closure to leaf-conditioned closure. It also separates memory created by eliminating an observable from the distinct closure problem posed by nonlinear networks.

1 Introduction

Let a scalar-output model be evaluated on a finite dataset $\{x_i\}_{i=1}^n$. Write

$$f(w) = \big(f(x_1; w), \ldots, f(x_n; w)\big)^\top \in \mathbb{R}^n, \qquad J(w) = D_w f(w) \in \mathbb{R}^{n \times p}, \tag{1}$$

and define the empirical tangent kernel

$$K(w) = J(w)J(w)^\top \in \mathbb{S}_+^n. \tag{2}$$

For a differentiable loss $L(w) = \ell(f(w))$, let $r(f) = \nabla_f \ell(f)$. Euclidean gradient flow gives

$$\dot{w} = -J^\top r, \qquad \dot{f} = -Kr, \qquad \dot{L} = -r^\top Kr \le 0. \tag{3}$$

At infinite width, the neural tangent kernel may become asymptotically constant, yielding a closed function-space description [Jacot et al., 2018, Lee et al., 2019]. At finite width, differentiating K generally introduces higher derivatives and leads to the neural tangent hierarchy [Huang and Yau, 2020].

The question here is exact and finite-dimensional. On a parameter domain $\mathcal{D}$, call (f, K) a *closed state* if there exists a single-valued map $\mathcal{F}$ such that

$$\dot{K}(w) = \mathcal{F}(f(w), K(w), r(f(w))) \qquad \text{for every } w \in \mathcal{D}. \tag{4}$$

Equivalently, the parameter-space vector field for $\dot{K}$ factors through the observation map $w \mapsto (f, K)$. This is stronger than having an equation along one trajectory and weaker than demanding an autonomous field on kernel space alone.

The main positive result is an exact closure for every bias-free two-factor linear network with scalar output. It is universal with respect to hidden width, input dimension, finite dataset, differentiable loss, and parameter state. It is also structurally degenerate: K drives f through (3), but $\dot{K}$ has no direct dependence on K. Once the prediction path is known, the kernel follows by quadrature. This one-sided structure permits exact elimination of K, producing a prediction-only equation with explicit memory.

The scalar deep-linear hierarchy then identifies the first failure of this property. Two factors close because the hierarchy terminates at the constant $e_0 = 1$. At three factors, the kernel velocity depends on e_1, which varies along regular fibers with fixed output and fixed tangent kernel. The missing information is not an active hidden state in this solvable model: squared-layer differences are conserved, and conditioning on the balancing leaf restores a Markovian closure.

Relation to prior work and contribution. The decomposition

$$K(x, x') = x^\top W^\top W x' + \|a\|^2 \, x^\top x'$$

appears explicitly in analyses of two-layer linear alignment, including the silent-alignment study of Atanasov et al. [2022]. Earlier work develops exact and asymptotic dynamics for two-factor and deep-linear networks, balancing laws, and structured mode-wise solutions [Saxe et al., 2014, Arora et al., 2018, Du et al., 2018, Xu and Ziyin, 2025]. This note places these ingredients in an exact state-reduction framework and establishes four linked results:

1. the full parameter vector field projects globally to (f, K) for arbitrary finite data and differentiable loss, without whitening, teacher, small-initialization, or mode-decoupling assumptions;
2. the projected law is independent of the nontrivial balancing leaf, yet is driven rather than kernel-autonomous;
3. exact elimination of the driven variable produces a non-fading Volterra representation of the same dynamics;
4. the first scalar deep-linear failure is characterized directly by the geometry of the fibers, and exact closure is recovered after conditioning on conserved leaf coordinates.

Together, these results identify a global projectability law at two factors and locate the first scalar factorization at which conserved leaf information becomes dynamically necessary.

2 Universal driven closure for two factors

Let $X \in \mathbb{R}^{n \times d_x}$ contain the data points as rows and consider

$$f = XW^\top a, \qquad W \in \mathbb{R}^{m \times d_x}, \qquad a \in \mathbb{R}^m. \tag{5}$$

Let $G = XX^\top$ be the fixed data Gram matrix and $r = \nabla_f \ell(f)$.

Proposition 2.1 (Universal two-factor tangent-kernel closure)**.** *For every hidden width m, input dimension d_x, finite dataset, differentiable loss, and parameter state,*

$$K = XW^\top WX^\top + \|a\|^2 G, \tag{6}$$

and gradient flow satisfies

$$\boxed{\dot{f} = -Kr, \qquad \dot{K} = -(Gr)f^\top - f(Gr)^\top - 2(f^\top r)G.} \tag{7}$$

Hence (f, K) is an exact closed state. The K-component depends only on (f, r) and the fixed data geometry G, not on the hidden factors or directly on K.

Proof. For sample i,

$$\nabla_W f_i = a x_i^\top, \qquad \nabla_a f_i = W x_i, \tag{8}$$

which gives (6). Put $s = X^\top r$. Gradient flow is

$$\dot{W} = -as^\top, \qquad \dot{a} = -Ws. \tag{9}$$

Because $G = XX^\top$ is fixed, $\dot{G} = 0$. Differentiating (6) gives

$$\dot{K} = X(\dot{W}^\top W + W^\top \dot{W})X^\top + \left(\frac{\mathrm{d}}{\mathrm{d}t}\|a\|^2\right) G. \tag{10}$$

Here the scalar derivative multiplying the fixed matrix G is

$$\frac{\mathrm{d}}{\mathrm{d}t}\|a\|^2 = 2\langle a, \dot{a}\rangle = -2a^\top Ws = -2f^\top r.$$

Substituting (9) into (10) therefore yields

$$\dot{K} = -(Xs)(a^\top WX^\top) - (XW^\top a)(Xs)^\top - 2(a^\top Ws)G. \tag{11}$$

Thus the last contribution is $(2\langle a, \dot{a}\rangle)G = -2(f^\top r)G$; it comes from differentiating $\|a\|^2$, never from differentiating the fixed matrix G. Since $Xs = Gr$ and $XW^\top a = f$, (7) follows. □

2.1 Driven closure rather than a constitutive self-law

Define

$$B_G(f, r) = -(Gr)f^\top - f(Gr)^\top - 2(f^\top r)G. \tag{12}$$

The exact reduced system is

$$\dot{f} = -Kr(f), \qquad \dot{K} = B_G(f, r(f)). \tag{13}$$

It is autonomous on the product state (f, K), but it is not a constitutive field $K \mapsto \dot{K}$ on the positive-semidefinite cone. The kernel has no instantaneous self-dynamics: its influence on its own future is indirect, through the prediction path that it helps generate. In this precise sense the closure is *driven*. Consequently, the exact case in which the hidden representation disappears is also a case in which a kernel-only gradient-flow question becomes vacuous unless one freezes the prediction, treats it as a control, or works on the product space.

2.2 A nontrivial foliation that the closure does not need

Proposition 2.2 (Balancing invariant and leaf independence)**.** *The matrix*

$$C = WW^\top - aa^\top \tag{14}$$

is conserved under (9). Nevertheless, the closed law (7) is independent of C.

Proof. Using (9),

$$\frac{\mathrm{d}}{\mathrm{d}t}(WW^\top) = -as^\top W^\top - Wsa^\top, \tag{15}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}(aa^\top) = -Wsa^\top - as^\top W^\top. \tag{16}$$

The derivatives are equal, so $\dot{C} = 0$. The formula (7) contains no C, proving leaf independence. □

This separates two notions that are often conflated. An invariant foliation may exist without its labels being required by a reduced state. The relevant question is not whether invariants exist, but whether the projected velocity changes across the leaves identified by the chosen observable.

2.3 Exact elimination and a non-fading memory law

The K-equation in (13) is integrable once the prediction path is known. Define

$$\mathcal{Q}_G(f, u) = (Gu)f^\top + f(Gu)^\top + 2(f^\top u)G = -B_G(f, u). \tag{17}$$

Proposition 2.3 (Exact prediction-only Volterra equation)**.** *Fix K_0 and write $r_t = r(f_t)$. The Markovian system (13) is equivalent to*

$$K_t = K_0 - \int_0^t \mathcal{Q}_G(f_s, r_s)\,\mathrm{d}s \tag{18}$$

and the closed prediction-only equation

$$\boxed{\dot{f}_t = -K_0 r_t + \int_0^t \mathcal{Q}_G(f_s, r_s) r_t\,\mathrm{d}s.} \tag{19}$$

Equivalently,

$$\begin{aligned}\dot{f}_t = {} & -K_0 r_t + \int_0^t \Big[(Gr_s)(f_s^\top r_t) + f_s(r_s^\top G r_t) \\ & \qquad\qquad + 2(f_s^\top r_s)Gr_t\Big]\,\mathrm{d}s.\end{aligned} \tag{20}$$

Proof. Integrating $\dot{K} = -\mathcal{Q}_G(f, r)$ gives (18). Substitution into $\dot{f}_t = -K_t r_t$ gives (19). Conversely, reconstructing K by (18) recovers both equations in (13). □

Equation (19) contains the complete past $s \in [0, t]$ without an explicit lag-decay factor. Old contributions may become small because the states (f_s, r_s) become small, but the representation has no structural fading kernel that would justify truncating the history to a short window. Conditional on K_0, there is no additional stochastic or orthogonal-dynamics remainder: the integral carries the entire effect of the eliminated kernel. The same deterministic training dynamics are therefore Markovian in (f, K) and history-dependent in f alone. Memory is a property of the chosen observable, not of the underlying vector field in isolation. This is an elementary, completely resolved instance of the projection principle emphasized by Mori–Zwanzig theory [Zwanzig, 1961, Mori, 1965, Chorin et al., 2002].

3 Scalar deep-linear hierarchy and failure beyond two factors

Consider the scalar width-one depth-d linear network

$$f(w) = \prod_{i=1}^{d} w_i \tag{21}$$

with scalar residual $r = \ell'(f)$. On a fixed sign component, put

$$q_i = w_i^2 > 0, \qquad e_k(q) = \sum_{1 \le i_1 < \cdots < i_k \le d} q_{i_1} \cdots q_{i_k}, \qquad e_0 = 1. \tag{22}$$

Then $f^2 = e_d$ and

$$K = \sum_{i=1}^{d} \left(\frac{\partial f}{\partial w_i} \right)^2 = e_{d-1}(q). \tag{23}$$

Gradient flow gives the common translation law

$$\dot{q}_i = -2fr \qquad \text{for every } i. \tag{24}$$

Proposition 3.1 (Finite symmetric-polynomial hierarchy)**.** *For $k = 1, \ldots, d$,*

$$\boxed{\dot{e}_k = -2fr(d-k+1)e_{k-1}.} \tag{25}$$

In particular,

$$\dot{f} = -e_{d-1}r = -Kr, \qquad \boxed{\dot{K} = -4fre_{d-2}.} \tag{26}$$

Thus $(f, e_1, \ldots, e_{d-1})$ is an exact finite-dimensional state.

Proof. Since every q_i has velocity $-2fr$,

$$\dot{e}_k = -2fr \sum_{i=1}^{d} \frac{\partial e_k}{\partial q_i}. \tag{27}$$

Every monomial of degree $k-1$ occurs in exactly $d-k+1$ of the derivatives, so

$$\sum_i \frac{\partial e_k}{\partial q_i} = (d-k+1)e_{k-1}. \tag{28}$$

The stated equations follow. □

At $d = 1$, $K \equiv 1$. At $d = 2$, (26) gives

$$\dot{f} = -Kr, \qquad \dot{K} = -4fr, \tag{29}$$

which is the scalar specialization of Proposition 2.1. From $d = 3$ onward, e_{d-2} is a new coordinate. The next theorem shows directly that it is not determined by (f, K) on generic fibers.

Theorem 3.2 (Regular-fiber obstruction from three factors onward)**.** *Let $d \ge 3$ and define*

$$\pi_d(q) = \big(e_d(q), e_{d-1}(q)\big), \qquad q \in \mathbb{R}^d_{>0}. \tag{30}$$

At every point for which the reciprocals $u_i = 1/q_i$ take at least three distinct values,

$$\operatorname{rank} D\big(e_d, e_{d-1}, e_{d-2}\big) = 3. \tag{31}$$

Consequently, the fiber of π_d is locally a smooth manifold of dimension $d-2$ containing curves along which e_{d-2} is nonconstant. For $d = 3$ the fiber itself is locally a curve, and e_1 varies along it.

Therefore, for any loss and output value with $fr \neq 0$, there is no leaf-independent single-valued law

$$\dot{K} = \mathcal{F}(f, K, r) \tag{32}$$

on the full positive state space. Three factors are the first scalar linear architecture at which the universal two-factor closure fails.

Proof. The change of variables $u_i = 1/q_i$ is a diffeomorphism of the positive orthant. Put

$$P = e_d(q), \qquad S = \frac{e_{d-1}(q)}{e_d(q)} = \sum_i u_i, \qquad T = \frac{e_{d-2}(q)}{e_d(q)} = \sum_{i<j} u_i u_j. \tag{33}$$

The transformations $(e_d, e_{d-1}, e_{d-2}) \leftrightarrow (\log P, S, T)$ are locally invertible for $P > 0$, so it suffices to study the differentials in u-coordinates:

$$\mathrm{d}\log P = -\sum_i \frac{\mathrm{d}u_i}{u_i}, \qquad \mathrm{d}S = \sum_i \mathrm{d}u_i, \qquad \mathrm{d}T = \sum_i (S - u_i)\mathrm{d}u_i. \tag{34}$$

The first two forms are independent unless all u_i are equal. Suppose that $\mathrm{d}T = A\,\mathrm{d}S + B\,\mathrm{d}\log P$. Comparing the coefficient of $\mathrm{d}u_i$ gives

$$S - u_i = A - \frac{B}{u_i}. \tag{35}$$

Hence every u_i is a root of the same quadratic

$$z^2 - (S - A)z - B = 0. \tag{36}$$

This is impossible when at least three distinct reciprocal values occur. The three differentials are therefore independent, proving (31).

The constant-rank theorem now makes the level set of (e_d, e_{d-1}) a local manifold of dimension $d-2$. Since $\mathrm{d}e_{d-2}$ is not in the span of $\mathrm{d}e_d$ and $\mathrm{d}e_{d-1}$, it has a nonzero derivative in some tangent direction to that fiber; a smooth curve tangent to that direction keeps (e_d, e_{d-1}) fixed while changing e_{d-2}. Along the positive sign component, fixed e_d means fixed f, and fixed e_{d-1} means fixed K. Equation (26) then gives different values of $\dot{K}$ whenever $fr \neq 0$. □

Remark (Explicit depth-three witness)**.** *For $d = 3$, take*

$$q^A = \left(\frac{2}{5}, \frac{5}{4}, 2\right), \qquad q^B = \left(\frac{5}{9}, \frac{3}{5}, 3\right).$$

Then $e_3(q^A) = e_3(q^B) = 1$ and $e_2(q^A) = e_2(q^B) = 19/5$, whereas

$$e_1(q^A) = \frac{73}{20}, \qquad e_1(q^B) = \frac{187}{45}.$$

Thus the two states have the same positive output $f = 1$ and the same tangent kernel $K = 19/5$, but (26) gives different kernel velocities whenever $r(1) \neq 0$.

The condition of three distinct reciprocal coordinates is open and dense. The obstruction is therefore not an isolated pair of parameter states but a generic local property of the positive deep-linear state space. It rules out a *universal* closure on that space. It does not rule out closure on a selected invariant leaf or on a more restrictive set of trajectories.

3.1 Exact closure on each balancing leaf

Equation (24) implies that

$$c_i = q_i - q_d, \qquad i = 1, \dots, d-1, \qquad \dot{c}_i = 0. \tag{37}$$

Fix a positive leaf c and write $q_d = s$, $q_i = s + c_i$. Define

$$\kappa_c(s) = e_{d-1}(s + c_1, \dots, s + c_{d-1}, s). \tag{38}$$

Proposition 3.3 (Leaf-conditioned exact closure)**.** *On every interval where all q_i are positive,*

$$\kappa_c'(s) = 2e_{d-2}(q(s,c)) > 0. \tag{39}$$

Hence $s = \kappa_c^{-1}(K)$ and

$$\boxed{\dot{K} = -4fr\, e_{d-2}\big(q(\kappa_c^{-1}(K), c)\big), \qquad \dot{c} = 0.} \tag{40}$$

Thus the failure in Theorem 3.2 is transverse to the balancing foliation: a conserved label, not temporal history, restores exact closure.

Proof. Differentiation under a common translation and Proposition 3.1 give

$$\frac{\mathrm{d}}{\mathrm{d}s} e_{d-1}(q(s,c)) = 2e_{d-2}(q(s,c)). \tag{41}$$

Positivity makes this derivative strictly positive. Inversion and (26) yield (40). □

3.2 Depth three in invariant coordinates

For $d = 3$, let

$$I(q) = \frac{1}{2}\sum_{i=1}^{3}(q_i - \bar{q})^2, \qquad \bar{q} = \frac{1}{3}e_1(q). \tag{42}$$

The common translation (24) gives $\dot{I} = 0$, and

$$I = \frac{e_1^2}{3} - e_2. \tag{43}$$

Since $K = e_2$ and $q_i > 0$,

$$e_1 = \sqrt{3(K + I)}. \tag{44}$$

The exact closed system is therefore

$$\boxed{\dot{f} = -Kr, \qquad \dot{K} = -4fr\sqrt{3(K+I)}, \qquad \dot{I} = 0.} \tag{45}$$

Because $f^2 = e_3$, $K = e_2$, and I recovers e_1, the state (f, K, I) is simply the complete symmetric state (e_3, e_2, e_1) written in invariant coordinates. The one-scalar augmentation is not a further compression of the hierarchy.

4 Interpretation and scope

4.1 What the two-factor theorem says

The strongest statement in this note is not that the kernel formula is difficult to derive. It is that, for two factors, the entire hidden representation can be quotiented out of the instantaneous joint dynamics for arbitrary data and loss. This quotient is universal across balancing leaves. The price is degeneracy: the reduced kernel equation is driven by prediction and contains no direct K-feedback. Thus exact closure and nontrivial kernel-space constitutive geometry are different questions. The present model answers the first and largely dissolves the second.

The Volterra representation sharpens the same point. Eliminating a known active variable can manufacture an exact memory equation without making the underlying system non-Markovian. Conversely, the deep-linear obstruction at three factors does not itself imply memory: the missing coordinate is conserved, and leaf conditioning suffices. Three distinct situations must therefore be kept separate:

$$\text{universal closure}, \qquad \text{closure after a static label}, \qquad \text{history required after elimination}. \tag{46}$$

The two-factor system realizes the first and third descriptions under different observables; scalar depth three realizes the second.

4.2 Reachable sets and nonlinear networks

The positive two-factor theorem holds on the full parameter space, so it automatically holds on every training-reachable subset. The negative result in Theorem 3.2 is an obstruction to a universal law on the full positive scalar deep-linear space. A narrower initialization manifold or a single invariant leaf may still admit closure; Proposition 3.3 gives this explicitly. The practically important balanced case is therefore worth naming. Exactly balanced initialization lies on $c = 0$, where all $q_i = s$, hence $K = ds^{d-1}$ and

$$\dot{K} = -2d(d-1)fr\left(\frac{K}{d}\right)^{\frac{d-2}{d-1}}.$$

At $d = 2$ this recovers $\dot{K} = -4fr$; from $d = 3$ onward, restricting to the balanced leaf restores the direct K-dependence absent from the universal two-factor law. Moreover, under the small-scale parameterization standard in deep-linear analyses [Saxe et al., 2014, Arora et al., 2018, Du et al., 2018], $w_i(0) = \varepsilon\xi_i$ with $\xi_i = O(1)$,

$$c_i(0) = q_i(0) - q_d(0) = \varepsilon^2(\xi_i^2 - \xi_d^2) \longrightarrow 0 \qquad (\varepsilon \to 0). \tag{47}$$

Thus small initialization starts near the balanced leaf in absolute scale. Since the c_i remain constant, whenever the common scale grows the relative imbalances $|q_i - q_j|/\bar{q}$ decrease even though their absolute values do not. The theorem limits global leaf-independent universality; by itself it does not establish practical nonclosure along balanced or near-balanced training trajectories.

For a smooth nonlinear network, the analogous projectability problem is structurally different: $\dot{K}$ contains Hessian contractions, and generic parameter-space fibers need not reflect the fibers actually reached from a standard initialization. A credible empirical projectability study must therefore sample states along real training trajectories, restrict vertical directions to the reachable set when possible, and scale the number of observations with model dimension. Generic Gaussian parameter states, a fixed two-point dataset, or a finite matched-pair separation cannot by themselves establish failure on the training-relevant manifold. Those diagnostics belong to a separate empirical study rather than to the present theorem note.

4.3 Limitations

The universal closure theorem concerns scalar-output, bias-free, two-factor linear networks. Input biases may be absorbed by homogeneous coordinates, whereas a separately trained output bias enlarges the state. The failure theorem is proved in the scalar width-one deep-linear subclass; it suffices to disprove a universal extension to all deeper factorizations, but it is not a classification of matrix-valued depth-three networks. The Volterra equation is exact after conditioning on K_0 and is nonconvolutional; projections with a nonzero orthogonal remainder fall outside this derivation. Discrete gradient descent, momentum, stochastic gradients, nonlinear activations, and Riemannian kernel geometry are separate extensions.

5 Conclusion

Two-factor linear networks provide an exact finite-width closure that is broader than the special regimes in which their learning dynamics are usually solved. For arbitrary finite data, differentiable loss, hidden width, and input dimension, the parameter flow projects to (f, K), independently of the balancing leaf. The resulting law is driven rather than kernel-autonomous, and exact elimination of K turns it into a full-history Volterra equation for f. The same dynamics are therefore Markovian or memory-bearing according to the observable retained.

The scalar deep-linear hierarchy pinpoints the first rupture. From three factors onward, generic fibers with fixed output and fixed tangent kernel carry varying e_{d-2} and hence varying kernel velocity. This failure is not irreducible memory: balancing labels restore closure, and at depth three the required invariant coordinate completes the symmetric hierarchy. The transition is thus from universal leaf-independent closure to leaf-conditioned closure. Whether analogous reductions hold on training-reachable sets of nonlinear networks is a separate empirical question.

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